

WEB APPLICATIONS WORKSHOP

QUIZ PACK

HTML • CSS • JAVASCRIPT

12 Topic Quizzes • 120 Questions • Detailed Answer Key

LEARN • PRACTISE • CHECK • IMPROVE

How to Use This Quiz Pack

Instructions: Answer each question by selecting the best option. Unless stated otherwise, each question has one correct answer. Complete the quiz questions before consulting the solutions section at the back of the pack.

Solutions: Each answer includes an explanation of the underlying concept so that the answer key can also be used for revision. Where useful, the explanation distinguishes the correct answer from common misconceptions represented by the other options.

Quiz 1 — HTML Basics

Select the best answer for each question.

1. **What is the main purpose of HTML in a webpage?**
 - A. To control presentation and layout
 - B. To structure and describe webpage content
 - C. To store data on the server
 - D. To compile JavaScript code
2. **Which declaration tells the browser that the document uses HTML5?**
 - A. `<html5>`
 - B. `<!DOCTYPE html>`
 - C. `<meta html="5">`
 - D. `<document type="html">`
3. **Which element contains metadata and resources that are not part of the main visible page content?**
 - A. `<body>`
 - B. `<main>`
 - C. `<head>`
 - D. `<section>`
4. **Which attribute specifies the language of an HTML document?**
 - A. `charset`
 - B. `lang`
 - C. `language`
 - D. `en`
5. **What is the purpose of the viewport meta element on responsive webpages?**
 - A. It sets the page background color
 - B. It controls how the page is sized and scaled on mobile devices
 - C. It loads the browser developer tools
 - D. It changes the HTML version
6. **Which attribute gives the destination URL of an `<a>` hyperlink?**
 - A. `src`
 - B. `alt`
 - C. `href`
 - D. `rel`
7. **What is the purpose of `<meta charset="UTF-8">`?**
 - A. To declare the document character encoding
 - B. To translate the page into English
 - C. To choose the page font
 - D. To set the browser window size

8. Which HTML creates a numbered list containing two items?

- A. `<ul><li>Soup</li><li>Salad</li></ul>`
- B. `<ol><p>Soup</p><p>Salad</p></ol>`
- C. `<list><item>Soup</item><item>Salad</item></list>`
- D. `<ol><li>Soup</li><li>Salad</li></ol>`

9. Which element gives a webpage its document title, commonly shown in the browser tab?

- A. `<h1>`
- B. `<title>`
- C. `<header>`
- D. `<caption>`

10. Which character reference displays a literal less-than sign in HTML text?

- A. `>`
- B. `&`
- C. `<`
- D. `"`

Quiz 2 — HTML Tables

Select the best answer for each question.

1. Which element creates a table row?
 - A. <td>
 - B. <tr>
 - C. <th>
 - D. <row>
2. Which element should normally be used for a header cell?
 - A. <thead>
 - B. <header>
 - C. <th>
 - D. <tdh>
3. What is the purpose of the <caption> element?
 - A. To merge table cells
 - B. To provide a title or description for a table
 - C. To create a table footer
 - D. To align the table
4. Which attribute allows a table cell to span more than one column?
 - A. rowspan
 - B. colspan
 - C. columns
 - D. span
5. Which grouping element contains the main body rows of a table?
 - A. <tbody>
 - B. <tablebody>
 - C. <main>
 - D. <body>
6. Which element represents an ordinary data cell in an HTML table?
 - A. <tr>
 - B. <thead>
 - C. <th>
 - D. <td>
7. What does rowspan="2" do when applied to a table cell?
 - A. Makes the cell span two rows
 - B. Creates two identical tables
 - C. Makes the cell span two columns
 - D. Adds two blank rows

- 8. In a simple table, what does scope="col" on a <th> identify?**
- A. The background color of the column
 - B. A cell that spans every column
 - C. A header for the cells in its column
 - D. The width of the column
- 9. Which element groups summary or totals rows at the foot of a table?**
- A. <footer>
 - B. <tfoot>
 - C. <bottom>
 - D. <summary>
- 10. Which content is most appropriate for an HTML table?**
- A. A product price list with product, quantity and price columns
 - B. The overall two-column layout of every webpage
 - C. A paragraph with one word highlighted
 - D. The main navigation menu solely to align its links

Quiz 3 — HTML Forms

Select the best answer for each question.

1. Which element is used to create an HTML form?
 - A. <input>
 - B. <form>
 - C. <fieldset>
 - D. <submit>
2. Why should a <label> normally be associated with a form control?
 - A. To make the control larger
 - B. To improve usability and accessibility by identifying the control
 - C. To make the control required
 - D. To submit the form automatically
3. Which input type is appropriate for an email address?
 - A. textmail
 - B. email
 - C. address
 - D. mailbox
4. What does the required attribute do?
 - A. Makes a field read-only
 - B. Prevents a form from being submitted while that field is empty
 - C. Automatically fills the field
 - D. Encrypts the field value
5. Which element is used when a user should choose one option from a drop-down list?
 - A. <select>
 - B. <choice>
 - C. <menuitem>
 - D. <optionlist>
6. Which attribute supplies the key used for an input value in submitted form data?
 - A. id
 - B. name
 - C. class
 - D. placeholder
7. Which label correctly associates with <input id="email" name="email">?
 - A. <label name="email">Email</label>
 - B. <label href="email">Email</label>
 - C. <label id="email">Email</label>
 - D. <label for="email">Email</label>

8. Which button type should be used for a form button that performs a scripted action without submitting the form?

- A. type="submit"
- B. type="reset"
- C. type="button"
- D. type="send"

9. How should radio inputs in the same form be grouped so that only one delivery option can be selected?

- A. Give them the same name and different values
- B. Give them the same id and no names
- C. Set required on every input but use different names
- D. Replace their type with checkbox

10. Why must a server validate form data even when the page uses required and type="email"?

- A. Because those attributes encrypt all submitted data
- B. Because browser validation automatically checks the database
- C. Because HTML forms can never validate input
- D. Because a client can bypass browser validation and send a request directly

Quiz 4 — HTML Containers

Select the best answer for each question.

1. What is the main purpose of the `<div>` element?
 - A. To create a generic block container for grouping content
 - B. To create a navigation landmark
 - C. To define the main heading
 - D. To create a table
2. Which element is a generic inline container?
 - A. `<section>`
 - B. `<span>`
 - C. `<div>`
 - D. `<article>`
3. Why are semantic elements generally preferred when their meaning matches the content?
 - A. They always load faster
 - B. They communicate the role or meaning of the content
 - C. They remove the need for CSS
 - D. Easier to read
4. Which element is most appropriate for the primary navigation links of a page?
 - A. `<div>`
 - B. `<nav>`
 - C. `<span>`
 - D. `<aside>`
5. Which element represents a thematic grouping of content, usually with a heading?
 - A. `<section>`
 - B. `<span>`
 - C. `<strong>`
 - D. `<br>`
6. Which element represents the dominant content of the page body?
 - A. `<aside>`
 - B. `<footer>`
 - C. `<main>`
 - D. `<span>`
7. Which element best fits a self-contained news story that could be reused independently?
 - A. `<article>`
 - B. `<span>`
 - C. `<nav>`
 - D. `<div>` with no semantic meaning

8. Which element is suitable for a sidebar of related information that is separate from the main discussion?

- A. `<main>`
- B. `<header>`
- C. `<section>` only because it is visually on the right
- D. `<aside>`

9. What is the role of a `<header>` inside an `<article>`?

- A. To contain document metadata such as `<title>`
- B. To group introductory content for that article
- C. To create the document's only permitted heading
- D. To force the article to the top of the browser window

10. What happens to the semantic meaning of a `<nav>` when CSS sets `display: flex` on it?

- A. It automatically becomes an `<article>`
- B. It loses its navigation role
- C. It keeps its navigation meaning while its layout changes
- D. Its links stop being interactive

Quiz 5 — Multimedia

Select the best answer for each question.

1. Which HTML element is used to embed video content?
 - A. <movie>
 - B. <media>
 - C. <video>
 - D. <embedvideo>
2. What does the controls attribute do on a <video> or <audio> element?
 - A. Adds browser-provided playback controls
 - B. Compresses the media file
 - C. Autoplays the media without restriction
 - D. Adds subtitles automatically
3. Which element can provide alternative media files for a <video> or <audio> element?
 - A. <file>
 - B. <source>
 - C. <mediafile>
 - D. <tracksrc>
4. Which element is used for timed text such as subtitles or captions with video?
 - A. <caption>
 - B. <subtitle>
 - C. <track>
 - D. <text>
5. What is an <iframe> commonly used for?
 - A. Embedding another HTML page or external web content within a page
 - B. Styling text
 - C. Creating an audio playlist
 - D. Defining metadata
6. Which attribute supplies a preview image for a <video> before playback?
 - A. preview
 - B. poster
 - C. thumbnail
 - D. alt
7. Which value of a <track> element's kind attribute identifies captions including relevant non-speech sounds?
 - A. chapters
 - B. metadata
 - C. subtitles
 - D. captions

8. Why might <video autoplay> fail to start playing automatically?

- A. The browser may restrict autoplay, especially when audio is audible
- B. The autoplay attribute permanently disables video
- C. Video can play only inside an <iframe>
- D. Autoplay always requires a table caption

9. What is appropriate alternative text for an image that is purely decorative and adds no information?

- A. Omit alt so the filename is always announced
- B. alt="image"
- C. alt=""
- D. Put a long caption in the src attribute

10. What does preload="metadata" request on an <audio> or <video> element?

- A. Download the entire file before displaying any page content
- B. Load information such as duration without a full media download being requested
- C. Start playback as soon as the HTML is parsed
- D. Generate a transcript automatically

Quiz 6 — CSS Basics

Select the best answer for each question.

1. What does CSS primarily control?

- A. The structure and meaning of HTML content
- B. The presentation and layout of webpages
- C. Server-side database queries
- D. The DNS lookup process

2. Which selector targets an element with id="menu"?

- A. .menu
- B. #menu
- C. menu
- D. *menu

3. Which selector targets all elements with class="card"?

- A. #card
- B. .card
- C. card
- D. @card

4. Which CSS property changes text color?

- A. font-color
- B. text-color
- C. color
- D. foreground

5. In the CSS box model, which area sits between the content and the border?

- A. margin
- B. padding
- C. outline
- D. gap

6. Which HTML correctly links an external stylesheet called styles.css?

- A. <link rel="stylesheet" href="styles.css">
- B. <style src="styles.css"></style>
- C. <script href="styles.css"></script>
- D. <css file="styles.css">

7. Which CSS rule correctly gives paragraphs a blue text color?

- A. p = blue;
- B. p { text-color: blue; }
- C. p: color(blue);
- D. p { color: blue; }

8. Two ordinary rules in the same stylesheet and cascade layer are `p { color: red; }` followed by `p { color: blue; }`. What color wins?

- A. Red, because the first rule always wins
- B. Blue, because the later rule breaks the tie
- C. Purple, because CSS mixes the values
- D. Neither, because repeated selectors are invalid

9. A box has `width: 200px`, `padding: 20px`, `border: 2px solid` and `box-sizing: border-box`. What is its total width including the border but excluding margin?

- A. 156px
- B. 240px
- C. 200px
- D. 244px

10. A paragraph has `color: green`. A child `<span>` has no color rule of its own. What normally happens?

- A. The span inherits the green text color
- B. The span must always be black
- C. The span becomes invisible
- D. The paragraph loses its color

Quiz 7 — CSS Layout

Select the best answer for each question.

1. Which CSS layout system is primarily one-dimensional?
 - A. Grid
 - B. Flexbox
 - C. Float
 - D. Table layout
2. Which CSS layout system is designed for rows and columns together?
 - A. Flexbox
 - B. Grid
 - C. Inline layout
 - D. Position static
3. Which declaration creates a flex container?
 - A. display: block
 - B. display: flex
 - C. position: flex
 - D. layout: flex
4. Which declaration creates a grid container?
 - A. display: grid
 - B. position: grid
 - C. layout: grid
 - D. grid: on
5. What is the purpose of a media query?
 - A. To play video files
 - B. To apply CSS conditionally based on characteristics such as viewport width
 - C. To load JavaScript modules
 - D. To validate HTML forms
6. Which Flexbox property aligns items along the main axis?
 - A. align-items
 - B. grid-template-columns
 - C. justify-content
 - D. position
7. Which declaration allows flex items to move onto additional lines when needed?
 - A. flex-direction: column
 - B. flex-wrap: wrap
 - C. overflow: hidden
 - D. white-space: nowrap

8. Which declaration defines three equal flexible column tracks in a grid?

- A. `grid-template-columns: 3px`
- B. `display: three-columns`
- C. `grid-template-rows: repeat(3, 1fr)`
- D. `grid-template-columns: repeat(3, 1fr)`

9. What does `gap: 16px` do on a grid container?

- A. Sets gutters between its rows and columns
- B. Adds 16px of padding inside each item
- C. Sets every item's width to 16px
- D. Adds a border around the grid

10. When does `@media (min-width: 768px)` apply?

- A. Only when the viewport is exactly 768 CSS pixels wide
- B. Whenever an image is at least 768 pixels wide
- C. When the viewport width is at least 768 CSS pixels
- D. Only when the physical screen has exactly 768 hardware pixels

Quiz 8 — JavaScript Basics

Select the best answer for each question.

1. What is JavaScript primarily used for in the browser?
 - A. Defining document structure
 - B. Adding logic and interactivity
 - C. Adding styling the document
 - D. Configuring the document
2. Which keyword declares a block-scoped variable whose value can be reassigned?
 - A. const
 - B. let
 - C. var
 - D. define
3. Which keyword declares a binding that cannot be reassigned?
 - A. const
 - B. let
 - C. static
 - D. fixed
4. Which of these is a JavaScript Boolean value?
 - A. "true"
 - B. TRUE
 - C. true
 - D. yes
5. What does the + operator do when both operands are numbers?
 - A. Concatenates them as text
 - B. Adds them
 - C. Compares them
 - D. Declares a variable
6. What is the result of 2 + "3" in JavaScript?
 - A. The number 5
 - B. The string "23"
 - C. The number 23
 - D. A syntax error
7. What is the result of 5 === "5"?
 - A. true
 - B. The string "true"
 - C. 5
 - D. false

8. Which expression converts the string "12" to the number 12?

- A. Number("12")
- B. String(12)
- C. "12" + "0"
- D. Boolean("12")

9. What does typeof 42 return?

- A. The number 42
- B. The string "integer"
- C. The string "number"
- D. The Boolean true

10. What does the expression 7 % 3 evaluate to?

- A. 2
- B. 1
- C. 3
- D. 0

Quiz 9 — Flow Control

Select the best answer for each question.

1. Which statement is used to run code conditionally?
 - A. if
 - B. repeat
 - C. define
 - D. switchon
2. What is the purpose of else in an if...else statement?
 - A. To repeat the if block
 - B. To provide code that runs when the if condition is false
 - C. To end the program
 - D. To declare a variable
3. Which loop is commonly used when you know how many iterations are required?
 - A. for
 - B. if
 - C. switch
 - D. try
4. What does break do inside a loop?
 - A. Restarts the loop
 - B. Stops the loop immediately
 - C. Skips only the current iteration
 - D. Creates a new loop
5. What does continue do inside a loop?
 - A. Ends the entire loop
 - B. Skips the rest of the current iteration and proceeds to the next
 - C. Returns a value from the loop
 - D. Pauses JavaScript
6. How many times does the body of for (let i = 0; i < 3; i++) run?
 - A. 0
 - B. 2
 - C. 4
 - D. 3
7. What happens when a while loop's condition is false before its first iteration?
 - A. Its body does not run
 - B. Its body runs exactly once
 - C. Its body runs forever
 - D. JavaScript automatically changes the condition to true

8. Which condition requires both isOpen and hasStock to be truthy before an order is accepted?

- A. isOpen || hasStock
- B. !isOpen
- C. isOpen && hasStock
- D. isOpen === hasStock

9. If the first branch of an if...else if...else chain runs, what happens to the later branches in that chain?

- A. They all run afterwards
- B. They are skipped
- C. Only the final else runs as well
- D. The first branch is repeated

10. In an ordinary switch statement, why is break commonly placed at the end of a case?

- A. To end the switch and prevent execution continuing into later cases
- B. To repeat the case until its value changes
- C. To skip evaluating the switch expression
- D. To make every case execute

Quiz 10 — Functions and Scope

Select the best answer for each question.

1. What is a function?

- A. A reusable block of code designed to perform a task
- B. A CSS selector
- C. An HTML container
- D. A browser storage area

2. What is a parameter?

- A. A value returned by a function
- B. A named input listed in a function definition
- C. A CSS property
- D. A loop counter only

3. What does return do in a function?

- A. Reloads the webpage
- B. Ends the function and optionally sends a value back to the caller
- C. Declares a global variable
- D. Repeats the function forever

4. What does block scope mean for a variable declared with let or const?

- A. It can be accessed from every script on the internet
- B. It is limited to the block in which it is declared and nested blocks
- C. It exists only inside HTML
- D. It is automatically global

5. What is hoisting in JavaScript?

- A. Moving HTML elements on the page
- B. The language behaviour in which some declarations are processed before execution reaches their written position
- C. Automatically importing libraries
- D. Reordering array items

6. Given function `total(price, quantity) { return price * quantity; }`, what are 4 and 3 in `total(4, 3)`?

- A. Return statements
- B. Global variables
- C. Arguments
- D. Function names

7. What is the result of calling function `greet() { console.log("Hello"); }`?

- A. It returns the string "Hello"
- B. It logs "Hello" and returns undefined
- C. It returns true
- D. It returns the console object

8. Given function `greet(name = "Guest") { return "Hello " + name; }`, what does `greet()` return?
- A. The string "Hello undefined"
 - B. The string "Guest"
 - C. undefined
 - D. The string "Hello Guest"
9. Inside a block with `let score = 20`, an outer scope already has `let score = 10`. Which value does `score` refer to inside that block after the inner declaration?
- A. 20, because the inner binding shadows the outer one
 - B. 10, because outer variables always take priority
 - C. 30, because the bindings are combined
 - D. A syntax error in every case
10. What happens if code reads a `let` variable in its scope before its declaration has been initialised?
- A. It always reads 0
 - B. It always reads undefined
 - C. It throws a `ReferenceError`
 - D. It automatically becomes a global variable

Quiz 11 — Objects

Select the best answer for each question.

- 1. What is a JavaScript object commonly used to represent?**
 - A. A collection of related data and behaviour
 - B. Only a single number
 - C. A CSS stylesheet
 - D. An HTML comment
- 2. In an object literal, what is a property?**
 - A. A named value associated with the object
 - B. A browser window
 - C. A loop statement
 - D. A CSS class
- 3. What is a method?**
 - A. A property whose value is a function
 - B. A table row
 - C. A primitive data type
 - D. A stylesheet rule
- 4. Which syntax accesses the name property of an object called restaurant?**
 - A. restaurant->name
 - B. restaurant.name
 - C. restaurant::name
 - D. restaurant(name)
- 5. What is a class in JavaScript?**
 - A. A template for creating objects with shared structure and behaviour
 - B. A CSS class selector
 - C. A database table only
 - D. An HTML section
- 6. Given `const field = "name"`, which expression reads that property from restaurant?**
 - A. restaurant.field
 - B. restaurant[field]
 - C. restaurant("field")
 - D. restaurant->field
- 7. Given `const order = { total: 10 }`, which statement can change the stored total?**
 - A. `order = { total: 12 };`
 - B. `const order = { total: 12 };` in the same scope
 - C. `order.total` cannot change because `const` freezes objects
 - D. `order.total = 12;`

8. Given `const first = { total: 10 }; const second = first; second.total = 20; what is first.total?`

- A. 20
- B. 10
- C. undefined
- D. A new empty object

9. What is the role of a constructor in a JavaScript class?

- A. To convert every property into a string
- B. To delete instances after each method call
- C. To initialise a new instance when the class is used with `new`
- D. To style HTML elements with a matching class attribute

10. Which expression creates an instance of a class named `MenuItem`?

- A. `class MenuItem()`
- B. `new MenuItem()`
- C. `MenuItem.new`
- D. `create class MenuItem`

Quiz 12 — Arrays, Events and the DOM

Select the best answer for each question.

1. What is an array used for?

- A. Storing an ordered collection of values
- B. Styling multiple elements
- C. Defining a webpage title
- D. Loading a stylesheet

2. Which method adds an item to the end of an array?

- A. add()
- B. push()
- C. appendItem()
- D. insertEnd()

3. What is the DOM?

- A. A programming interface that represents an HTML document as objects that JavaScript can work with
- B. A CSS layout model
- C. A network protocol
- D. A database format

4. Which method attaches an event handler without replacing existing handlers registered the same way?

- A. addEventListener()
- B. setEvent()
- C. onEvent()
- D. listen()

5. In an event listener, what does event.currentTarget refer to?

- A. The element on which the listener is registered
- B. The browser address bar
- C. The first element in the document
- D. The array currently being processed

6. Given `const items = ["Soup", "Salad", "Pasta"]`, what does `items[1]` return?

- A. "Soup"
- B. "Pasta"
- C. undefined
- D. "Salad"

7. Which array method creates a new array containing values that pass a test?

- A. filter()
- B. push()
- C. pop()
- D. reverse()

- 8. What does `document.querySelector(".card")` return when several elements match?**
- A. An array of every matching element
 - B. The last matching element
 - C. The first matching element in document order
 - D. The text of every matching element
- 9. Which property should be assigned to replace an element's contents with plain text without parsing that text as HTML?**
- A. `innerHTML`
 - B. `textContent`
 - C. `className`
 - D. `style`
- 10. In a non-passive submit listener, what does `event.preventDefault()` do when the event is cancelable?**
- A. Cancels the event's default form-submission action
 - B. Removes every listener from the form
 - C. Stops all other JavaScript on the page
 - D. Always prevents the event from bubbling

SOLUTIONS & ANSWER KEY

Detailed explanations for all 120 questions

Quiz 1 — HTML Basics — Solutions

1. B — To structure and describe webpage content

Explanation: HTML defines the structure and meaning of the content in a webpage. Elements such as headings, paragraphs, links, images and forms describe what the content is and how it is organised. CSS is used for presentation and layout, while JavaScript supplies programming logic and interactivity.

2. B — `<!DOCTYPE html>`

Explanation: The declaration `<!DOCTYPE html>` tells the browser to interpret the document using the modern HTML standard and standards mode. It appears before the `<html>` element and is a declaration rather than an ordinary HTML element. The other options are not valid HTML5 doctype syntax.

3. C — `<head>`

Explanation: The `<head>` contains document metadata and references to resources rather than the page's main visible content. It commonly contains the `<title>` element, meta elements and links to stylesheets. The `<body>` contains the content that is rendered as the main page interface.

4. B — `lang`

Explanation: The `lang` attribute identifies the natural language of the document, for example `<html lang="en">` for English. This helps browsers, search engines and assistive technologies interpret the content correctly. `charset` specifies character encoding, not the document language.

5. B — It controls how the page is sized and scaled on mobile devices

Explanation: The viewport meta element tells mobile browsers how the page should relate to the device viewport. A common setting is `width=device-width, initial-scale=1.0`, which helps responsive layouts use the available screen width correctly. It does not itself create a responsive design; the CSS still needs to adapt the layout.

6. C — `href`

Explanation: The `href` attribute supplies the link destination, such as `href="menu.html"`. The link text appears between the opening and closing `<a>` tags. `src` identifies a resource to embed, while `rel` describes a relationship to a linked resource; neither replaces `href`.

7. A — To declare the document character encoding

Explanation: This declaration identifies UTF-8 as the character encoding used to interpret the document bytes. Correct encoding helps characters such as accented letters display properly. It does not translate content or select a font; the `lang` attribute identifies the natural language.

8. D — `<ol><li>Soup</li><li>Salad</li></ol>`

Explanation: An `<ol>` represents an ordered list, and each item belongs in an `<li>` element. Browsers normally display numbers for this list. A `<ul>` represents an unordered list, usually shown with bullets; paragraphs alone do not provide the required list-item structure.

9. B — `<title>`

Explanation: The `<title>` element belongs in the `<head>` and names the document for browser tabs, bookmarks and similar uses. A visible page heading belongs in the `<body>`, often as an `<h1>`. The document title and the main heading serve related but distinct purposes.

10. C — `<`

Explanation: The character reference `<` produces the character `<` without starting an HTML tag. For example, `<p>` displays the text `<p>`. The alternatives represent a greater-than sign, an ampersand and a double quotation mark respectively.

Quiz 2 — HTML Tables — Solutions

1. B — `<tr>`

Explanation: The `<tr>` element represents one row in a table. Header cells and data cells are placed inside that row using `<th>` and `<td>` elements. The `<td>` and `<th>` elements create cells, not rows.

2. C — `<th>`

Explanation: The `<th>` element represents a header cell and gives the cell header semantics. This is useful not only visually but also for accessibility because assistive technologies can associate headers with related data cells. `<thead>` groups header rows; it is not itself a cell.

3. B — To provide a title or description for a table

Explanation: The `<caption>` element provides a title or short description for the table as a whole. It is placed as the first child of the `<table>` element and helps users understand what the table represents. It is not used to merge cells or position the table.

4. B — `colspan`

Explanation: The `colspan` attribute specifies the number of columns that a single table cell should span. For example, `colspan="2"` makes one cell occupy the space of two columns. `rowspan` performs the corresponding operation vertically across rows.

5. A — `<tbody>`

Explanation: The `<tbody>` element groups the primary body rows of a table. It sits between optional `<thead>` and `<tfoot>` groups and helps organise the table structurally. The ordinary page `<body>` element has a completely different purpose and cannot replace `<tbody>`.

6. D — `<td>`

Explanation: A `<td>` contains a data value, such as a product price, inside a table row. A `<th>` identifies a header for related cells. A `<tr>` groups cells into a row, while `<thead>` groups the table header rows.

7. A — Makes the cell span two rows

Explanation: `rowspan="2"` makes one cell occupy a position across two rows. The remaining cells must be arranged to account for that occupied space. `colspan` extends a cell horizontally across columns; `rowspan` does not duplicate the cell content or create additional rows by itself.

8. C — A header for the cells in its column

Explanation: `scope="col"` explicitly identifies the header as applying to its column, helping establish the relationship between header and data cells. For example, a Price header can label the prices below it. `scope="row"` instead identifies a row header; `scope` is not a visual styling attribute.

9. B — `<tfoot>`

Explanation: The `<tfoot>` element groups table footer rows, which may contain totals or other summaries. It contains `<tr>` rows just as `<thead>` and `<tbody>` do. A page or section `<footer>` has a different semantic role and does not replace the table-specific grouping element.

10. A — A product price list with product, quantity and price columns

Explanation: A table is appropriate when values have meaningful relationships across rows and columns, as in a product price list. Those relationships justify header and data cells. General page arrangement should use CSS layout, such as Flexbox or Grid, rather than table markup solely for positioning.

Quiz 3 — HTML Forms — Solutions

1. B — <form>

Explanation: The <form> element groups controls that collect data from a user and defines the context in which those values are submitted. Inputs, selects, textareas and buttons normally sit inside a form. An <input> is an individual control, not the form container itself.

2. B — To improve usability and accessibility by identifying the control

Explanation: A <label> gives a form control an explicit text description. When it is correctly associated with the control, users can often click the label to focus or activate the control, and assistive technologies can announce the control's purpose. A label does not make a field required; that is the job of validation attributes such as required.

3. B — email

Explanation: type="email" identifies the expected value as an email address. Browsers can apply email-specific constraint validation and may provide a more appropriate keyboard on mobile devices. It is still important to validate submitted data on the server as well when the form is processed by a back end.

4. B — Prevents a form from being submitted while that field is empty

Explanation: The required attribute marks a control as one that must contain a valid value before the browser will consider the form valid for normal submission. It participates in HTML constraint validation. It does not make the control read-only or automatically supply a value.

5. A — <select>

Explanation: The <select> element creates a list from which the user can choose an option, with individual choices defined by <option> elements. Depending on its attributes it can present a drop-down list or a larger list box. The other proposed element names are not the standard HTML control for this purpose.

6. B — name

Explanation: The name attribute provides the key for a submitted control value, for example name="email". An id can connect a label to the control or identify it for scripting, but it does not replace name. A text input without a name is normally omitted from form submission.

7. D — <label for="email">Email</label>

Explanation: A label's for attribute must match the id of the control it identifies. This creates an explicit association for assistive technologies and lets users activate or focus the control through its label. Repeating the input's id on the label would create duplicate IDs, not this association.

8. C — type="button"

Explanation: type="button" gives the button no default form-submission action, so a click handler can perform the intended task. A submit button initiates submission, and a reset button resets controls. Explicitly choosing the type avoids accidental submission from a button placed inside a form.

9. A — Give them the same name and different values

Explanation: Radio inputs in the same form and tree form a group when they share a non-empty name. Selecting one then deselects another in that group. Different values identify which option was selected for submission; each control should still have its own unique id for labelling.

10. D — Because a client can bypass browser validation and send a request directly

Explanation: Browser validation provides immediate feedback, but the server cannot assume every request came through that interface. A client can modify the page or send a request directly. The server therefore needs its own checks for acceptable values and business rules before processing or storing submitted data.

Quiz 4 — HTML Containers — Solutions

1. A — To create a generic block container for grouping content

Explanation: The `<div>` element is a generic block-level container used to group flow content when no more meaningful semantic element is appropriate. It is frequently used as a styling or scripting hook. It carries no special meaning about the content by itself.

2. B — `<span>`

Explanation: The `<span>` element is a generic inline container. It is useful for grouping a short piece of text or other phrasing content so that CSS or JavaScript can target it without introducing a block-level semantic section. A `<div>` is the corresponding generic block container.

3. B — They communicate the role or meaning of the content

Explanation: Semantic elements communicate what a region of content represents, for example navigation, an article, a section or a footer. This can make document structure clearer for developers and assistive technologies. They do not eliminate CSS or JavaScript and they are not inherently faster simply because they are semantic.

4. B — `<nav>`

Explanation: The `<nav>` element represents a section whose purpose is navigation, normally containing important navigation links. Using `<nav>` communicates that role explicitly instead of placing the links inside a generic `<div>`. Not every group of links must be in `<nav>`, but primary or significant navigation is a good fit.

5. A — `<section>`

Explanation: The `<section>` element represents a thematic grouping of content, normally identified by a heading. It is appropriate when the grouped material forms a meaningful section of the document. A `<span>` is an inline generic container and does not express this structural meaning.

6. C — `<main>`

Explanation: The `<main>` element identifies the page's dominant content, such as the article or application area central to its purpose. Repeated navigation and site-wide footer information normally sit outside it. This semantic region also helps assistive-technology users navigate directly to the principal content.

7. A — `<article>`

Explanation: An `<article>` represents a self-contained composition, such as a news story, blog post or forum entry, that could stand on its own. A `<section>` groups material by theme, while a generic `<div>` only groups content. Choose the element according to the content's meaning.

8. D — `<aside>`

Explanation: An `<aside>` represents content tangentially related to the surrounding material, such as related reading or a contextual sidebar. Its meaning comes from that relationship, not its position on the screen. CSS can place it beside or below the main content without changing its semantics.

9. B — To group introductory content for that article

Explanation: A `<header>` can introduce its containing article with a heading, author information or other introductory material. It is distinct from `<head>`, which holds document metadata. A document may have headers for different sections; the element does not fix content to the top of the viewport.

10. C — It keeps its navigation meaning while its layout changes

Explanation: display: flex changes how the navigation container lays out its children. The HTML element remains a <nav> and still identifies a navigation region. This illustrates the separate responsibilities of semantic HTML and CSS presentation: a layout choice does not rename the element or remove its links.

Quiz 5 — Multimedia — Solutions

1. C — <video>

Explanation: The <video> element is the native HTML element for embedding and playing video. It can reference a source directly or contain one or more <source> elements. Browser controls, poster images, captions and other playback behaviour can be configured with attributes and child elements.

2. A — Adds browser-provided playback controls

Explanation: The controls attribute asks the browser to display its built-in playback interface, typically including play/pause, seek, volume and fullscreen controls where supported. It does not compress the media or create subtitles. Autoplay is controlled separately and is also subject to browser autoplay policies.

3. B — <source>

Explanation: A <source> element lets you supply alternative media resources inside <video> or <audio>. The browser can examine the listed resources and use one it supports. This is useful when the same media is available in more than one format or encoding.

4. C — <track>

Explanation: The <track> element attaches timed text data to audio or video, including captions, subtitles, descriptions or chapter information depending on the kind attribute. A WebVTT file is commonly used as the timed-text source. The table <caption> element is unrelated to multimedia captions.

5. A — Embedding another HTML page or external web content within a page

Explanation: An <iframe> embeds another browsing context inside the current webpage. It is often used for content such as maps, hosted videos or another webpage supplied by an external service. Because the embedded content is a separate document, security and permission restrictions may apply.

6. B — poster

Explanation: The poster attribute gives the URL of an image the browser can show while video content is not yet available for display, commonly before playback. It does not generate captions or describe the video for screen readers. Caption tracks and other accessible alternatives need separate provision.

7. D — captions

Explanation: kind="captions" identifies timed text that can convey speech and meaningful sounds such as a doorbell or laughter. This supports viewers who cannot hear the audio. Subtitles typically translate or transcribe speech for viewers who can hear the soundtrack; chapters identify navigation points.

8. A — The browser may restrict autoplay, especially when audio is audible

Explanation: Autoplay is a request, not a guarantee. Browsers may require user interaction or apply other restrictions, particularly for audible media. A page should therefore provide a usable way to start playback instead of assuming that adding autoplay will always cause the video to play.

9. C — alt=""

Explanation: An empty alt attribute signals that the image contributes no information that needs a text alternative, allowing assistive technologies to skip it in normal reading. Omitting alt is different and may lead to an unhelpful filename being announced. Informative images instead need meaningful alternative text.

10. B — Load information such as duration without a full media download being requested

Explanation: preload="metadata" is a hint that metadata such as duration may be fetched before playback rather than requesting the whole resource. The browser decides how to honour the hint, so it is not an exact network guarantee. It does not enable autoplay or produce captions or a transcript.

Quiz 6 — CSS Basics — Solutions

1. B — The presentation and layout of webpages

Explanation: CSS controls how HTML content is presented, including colors, typography, spacing, borders and layout. HTML provides the structure and semantics; CSS applies visual presentation to that structure. CSS does not perform server-side database operations or DNS resolution.

2. B — #menu

Explanation: The # selector targets an element by its id value, so #menu targets an element whose id is menu. An ID should normally be unique within a document. A leading dot selects a class instead.

3. B — .card

Explanation: The . selector targets elements by class, so .card matches every element whose class list contains card. Unlike an ID, the same class can be applied to many elements, which makes classes particularly useful for reusable styling.

4. C — color

Explanation: The color property sets the foreground color of text and can also affect other foreground features that use `currentColor`. CSS does not use properties called `font-color` or `text-color` for this purpose. The standard property name is simply `color`.

5. B — padding

Explanation: Padding is the space between an element's content box and its border. Margin lies outside the border and separates the element from neighbouring elements. Remembering the order content → padding → border → margin is a useful way to visualise the CSS box model.

6. A — <link rel="stylesheet" href="styles.css">

Explanation: A <link> element with `rel="stylesheet"` identifies an external CSS resource, and `href` supplies its URL. It normally belongs in the document head. A <style> element instead contains CSS rules directly; a script element is not the mechanism for attaching a CSS stylesheet.

7. D — p { color: blue; }

Explanation: A CSS rule consists of a selector followed by a declaration block in braces. Each declaration uses a property, a colon and a value, normally followed by a semicolon. Here `p` selects paragraphs and `color: blue` sets their foreground text color.

8. B — Blue, because the later rule breaks the tie

Explanation: These rules have the same origin, importance, layer and selector specificity, so their order determines the result: the later blue declaration wins. CSS does not mix the colors. Source order resolves a tie; it does not automatically override a declaration with higher cascade priority.

9. C — 200px

Explanation: With border-box sizing, the specified 200px width includes the content, left and right padding, and borders. The content width is therefore $200 - 40 - 4 = 156\text{px}$. With content-box sizing the same declared width would apply only to the content, giving a 244px outer width.

10. A — The span inherits the green text color

Explanation: The `color` property is inherited, so the `span` normally uses its parent's computed text color when no declaration sets a different value. This avoids repeating text styling on every descendant. Not every CSS property is inherited; properties such as `margin` do not normally pass down this way.

Quiz 7 — CSS Layout — Solutions

1. B — Flexbox

Explanation: Flexbox is primarily a one-dimensional layout system: it arranges items along a main axis, either a row or a column, while also providing control over alignment on the cross axis. It is especially useful for navigation bars, toolbars and other layouts dominated by one direction.

2. B — Grid

Explanation: CSS Grid is designed for two-dimensional layouts in which rows and columns are considered together. It lets you define grid tracks, place items into cells or named areas and control gaps and alignment. Flexbox and Grid often complement one another rather than competing as mutually exclusive choices.

3. B — display: flex

Explanation: display: flex turns an element into a flex container and its direct children become flex items. Flexbox-related properties can then control direction, wrapping, alignment and distribution. position: flex and layout: flex are not valid ways to establish a flex formatting context.

4. A — display: grid

Explanation: display: grid establishes a grid formatting context and makes the element's direct children grid items. You can then define rows and columns with properties such as grid-template-columns and grid-template-rows. The alternative declarations shown are not valid CSS for creating a grid container.

5. B — To apply CSS conditionally based on characteristics such as viewport width

Explanation: A media query applies a group of CSS rules only when specified media conditions are satisfied. Responsive layouts commonly use conditions based on viewport width, but queries can also examine characteristics such as orientation or user preferences. Media queries do not play media files despite the word 'media' in their name.

6. C — justify-content

Explanation: justify-content distributes available space and aligns flex items along the main axis. The direction of that axis depends on flex-direction. align-items works on the cross axis, so it should not be treated as an interchangeable name for horizontal or vertical alignment.

7. B — flex-wrap: wrap

Explanation: flex-wrap: wrap allows the flex container to use multiple flex lines when its items cannot fit on one line under the layout rules. The default is nowrap. Changing flex-direction changes the main axis, while hiding overflow clips content rather than enabling line wrapping.

8. D — grid-template-columns: repeat(3, 1fr)

Explanation: repeat(3, 1fr) creates three column tracks with equal flex factors, sharing available space subject to track sizing and content constraints. grid-template-rows defines rows instead. Long unbreakable content can still affect sizing, so equal fractions alone are not a guarantee against content overflow.

9. A — Sets gutters between its rows and columns

Explanation: gap sets the spacing between grid tracks; one value applies to both row and column gaps. It does not add padding inside each item or an outer border. The container's padding and margins control different spaces and may be used alongside gaps.

10. C — When the viewport width is at least 768 CSS pixels

Explanation: min-width is an inclusive lower bound, so this query matches a viewport width of 768 CSS pixels or more. It tests the viewport, not an individual image or the hardware pixel count. Responsive breakpoints let selected styles change when the available layout space reaches a threshold.

Quiz 8 — JavaScript Basics — Solutions

1. B — Adding programming logic and interactivity

Explanation: In the browser, JavaScript provides programming logic that can respond to events, perform calculations, change the DOM and coordinate interactive behaviour. HTML supplies structure and CSS supplies presentation. JavaScript can also be used outside the browser, but this question is specifically about its browser role.

2. B — let

Explanation: let declares a block-scoped binding that may later be assigned a different value. It is appropriate when the variable genuinely needs reassignment. const is also block-scoped but prevents reassignment of the binding after it has been initialised.

3. A — const

Explanation: const declares a block-scoped binding that cannot be reassigned to another value after declaration. This does not make an object or array deeply immutable; its contents may still be changed unless additional measures are used. The restriction applies to reassignment of the binding itself.

4. C — true

Explanation: true is one of JavaScript's two Boolean literals; the other is false. Quoting "true" produces a string rather than a Boolean, and JavaScript literals are case-sensitive, so TRUE is not the Boolean literal true.

5. B — Adds them

Explanation: When both operands are numeric, the + operator performs arithmetic addition. JavaScript also uses + for string concatenation, so the result depends on operand types. For example, 2 + 3 produces 5, whereas "2" + 3 produces the string "23" because one operand is a string.

6. B — The string "23"

Explanation: Because one operand is a string, + converts the other operand to a string and concatenates them. The result is "23", not numeric addition. If arithmetic is intended, an explicit conversion such as 2 + Number("3") produces the number 5.

7. D — false

Explanation: Strict equality compares these values without converting the string to a number. The operands have different types, so 5 === "5" is false. This differs from loose equality, ==, which can perform type conversion and would consider this particular pair equal.

8. A — Number("12")

Explanation: Number("12") converts this numeric string into the number 12, which can then be used in arithmetic. String(12) performs the opposite type of conversion. Concatenating "0" produces "120", while Boolean("12") produces true because a non-empty string is truthy.

9. C — The string "number"

Explanation: typeof returns a string describing a value's type, and an ordinary numeric literal such as 42 has type "number". JavaScript does not use "integer" as a typeof result. The result of typeof is itself a string, even when the value being examined is numeric.

10. B — 1

Explanation: The % operator calculates a remainder. Dividing 7 by 3 gives two whole groups of 3 and leaves 1, so the result is 1. It does not return the quotient or a percentage; remainder calculations are often useful for checking even and odd integer values.

Quiz 9 — Flow Control — Solutions

1. A — if

Explanation: An if statement evaluates a condition and executes its associated block when that condition is truthy. It is one of the fundamental ways to introduce selection into a program. Other branches can be added with else if and else.

2. B — To provide code that runs when the if condition is false

Explanation: else supplies an alternative block that executes when the preceding if condition is not satisfied. This lets a program choose between two paths. It does not repeat the if block and it does not terminate the entire program.

3. A — for

Explanation: A for loop is commonly used when iteration follows a counted pattern, because its syntax groups initialisation, a continuation condition and an update expression in one place. Other loops such as while may be better when the number of repetitions is not naturally expressed as a count.

4. B — Stops the loop immediately

Explanation: break immediately exits the nearest enclosing loop or switch statement and continues execution after it. This is useful when the program has found what it needs and no further iterations are required. It is different from continue, which keeps the loop running.

5. B — Skips the rest of the current iteration and proceeds to the next

Explanation: continue skips the remainder of the current loop iteration and proceeds to the next iteration. The loop itself is not terminated. In a for loop, the update expression is then processed before the next condition check.

6. D — 3

Explanation: The loop body runs with i equal to 0, 1 and 2. After the third iteration, i becomes 3 and the condition $i < 3$ is false. Counting the actual values that satisfy the condition helps avoid off-by-one errors when writing loops.

7. A — Its body does not run

Explanation: A while loop checks its condition before entering the body, so an initially false condition causes zero iterations. A do...while loop checks after the body and therefore runs at least once. Choosing the appropriate loop depends on whether the first execution must occur unconditionally.

8. C — isOpen && hasStock

Explanation: Logical AND, &&, is truthy only when both operands are truthy. In an if condition, isOpen && hasStock therefore requires both checks to succeed. Logical OR accepts either truthy operand, while equality could be true when both values are false and would not express this requirement.

9. B — They are skipped

Explanation: An if...else if...else chain selects the first branch whose condition is truthy, or the final else if none match. Once a branch is selected, later branches in that same chain are skipped. Separate if statements are different: more than one of them may execute.

10. A — To end the switch and prevent execution continuing into later cases

Explanation: After a matching case is found, execution continues through subsequent statements unless control exits. break exits the switch and prevents unintended fall-through into later cases. Omitting break can be deliberate, but it must match the intended behaviour; each case is not automatically isolated.

Quiz 10 — Functions and Scope — Solutions

1. A — A reusable block of code designed to perform a task

Explanation: A function groups statements into a reusable unit that can be invoked when needed. Functions can receive inputs through parameters, perform work and optionally return a result. They help avoid duplicated code and make larger programs easier to structure.

2. B — A named input listed in a function definition

Explanation: A parameter is a named placeholder written in a function definition. When the function is called, argument values are supplied and become available through those parameters. Parameters therefore describe the inputs the function expects; arguments are the actual values passed at a particular call.

3. B — Ends the function and optionally sends a value back to the caller

Explanation: `return` ends the current function call and can send a value back to the code that called the function. Statements after an executed `return` in the same function call are not run. A function with no explicit returned value produces `undefined` as its result.

4. B — It is limited to the block in which it is declared and nested blocks

Explanation: Block scope means a `let` or `const` binding is available within the block where it is declared, including nested blocks, but not outside that block. Blocks are commonly delimited by braces in structures such as `if` statements, loops and functions. This helps prevent variables from unintentionally leaking into wider scopes.

5. B — The language behaviour in which some declarations are processed before execution reaches their written position

Explanation: Hoisting describes aspects of how JavaScript declarations are created and made known to a scope before execution reaches the line where they appear. Different declarations behave differently: function declarations are callable before their written position, while `let` and `const` are not usable before initialisation and are subject to the temporal dead zone. It is therefore misleading to think of hoisting as JavaScript literally moving source code.

6. C — Arguments

Explanation: Arguments are the actual values supplied in a function call, so 4 and 3 are arguments here. `price` and `quantity` are the parameters named in the definition. During this call those parameters receive the argument values, and the function returns 12.

7. B — It logs "Hello" and returns undefined

Explanation: `console.log` displays a message, but displaying output is not the same as returning a value to the caller. This function reaches its end without a `return` statement, so its result is `undefined`. To return the greeting string, it would need `return "Hello"`.

8. D — The string "Hello Guest"

Explanation: The default parameter value `"Guest"` is used because the call supplies no value for `name`. The `return` expression then concatenates the greeting and that value. A default also applies to an explicit `undefined` argument, but not automatically to `null` or an empty string.

9. A — 20, because the inner binding shadows the outer one

Explanation: The nearest applicable lexical binding is used, so the inner `score` shadows the outer `score` within that block. The two variables are separate; declaring the inner one does not overwrite the outer value. Outside the block, the outer `score` still refers to 10.

10. C — It throws a **ReferenceError**

Explanation: A `let` binding exists in its scope before its declaration is executed, but it cannot be read while uninitialised. This interval is called the temporal dead zone. Reading it then throws a `ReferenceError`; the behaviour differs from a `var` binding initialised to `undefined`.

Quiz 11 — Objects — Solutions

1. A — A collection of related data and behaviour

Explanation: A JavaScript object groups related values as properties and can also group behaviour as methods. This makes objects useful for representing entities such as a restaurant, order, user or product. They are reference values rather than primitive values.

2. A — A named value associated with the object

Explanation: A property is a named entry on an object that stores or refers to a value. For example, a restaurant object might have name, address and isOpen properties. Properties can be accessed with dot notation or bracket notation when appropriate.

3. A — A property whose value is a function

Explanation: A method is a function associated with an object, normally stored as a property whose value is callable. Calling a method expresses behaviour related to that object, such as `order.calculateTotal()`. This distinguishes methods from ordinary data properties such as name or price.

4. B — `restaurant.name`

Explanation: `restaurant.name` uses dot notation to access the property named name on the restaurant object. Bracket notation, `restaurant["name"]`, could also access the same property. The arrow, double-colon and function-call forms shown in the alternatives are not normal JavaScript property access syntax.

5. A — A template for creating objects with shared structure and behaviour

Explanation: A JavaScript class defines a reusable pattern for constructing objects that share behaviour and an initial structure. Calling the class with `new` invokes its constructor and creates an instance. Classes are built on JavaScript's prototype-based object model, so they are not the same thing as CSS classes despite the shared word 'class'.

6. B — `restaurant[field]`

Explanation: Bracket notation evaluates field and uses its value, "name", as the property key. `restaurant[field]` therefore reads `restaurant.name` in this example. Dot notation `restaurant.field` would look for a property literally named field, which is a different key.

7. D — `order.total = 12;`

Explanation: `const` prevents reassignment of the order binding, but it does not freeze the ordinary object referenced by that binding. Assigning `order.total = 12` changes the existing property. Reassigning order would fail, and redeclaring the same `const` name in the same scope is a syntax error.

8. A — 20

Explanation: Both variables refer to the same object. Assigning first to second copies the reference rather than cloning the object, so changing the total through second is also visible through first. This behaviour is central to understanding object sharing and unintended mutations.

9. C — To initialise a new instance when the class is used with `new`

Explanation: The constructor performs initialisation when an instance is created with `new`, commonly assigning starting properties through `this`. For example, a `Product` constructor may store a name and price. It belongs to the JavaScript object model and is unrelated to selecting HTML elements with CSS classes.

10. B — new MenuItem()

Explanation: new MenuItem() invokes the class constructor and creates an instance, which can access the class's instance methods. Constructor arguments can be supplied inside the parentheses when needed. A class declaration defines the class; it does not by itself create the individual menu-item instances.

Quiz 12 — Arrays, Events and the DOM — Solutions

1. A — Storing an ordered collection of values

Explanation: An array stores an ordered collection of values under one variable. Each element has a numeric index beginning at 0, and arrays provide methods for adding, removing, searching and transforming values. The values do not all have to be of the same JavaScript type.

2. B — push()

Explanation: push() appends one or more values to the end of an array and returns the array's new length. For example, items.push("Soup") adds "Soup" after the existing elements. Methods such as add() or appendItem() are not standard Array methods.

3. A — A programming interface that represents an HTML document as objects that JavaScript can work with

Explanation: The DOM, or Document Object Model, is the programming interface through which the browser represents an HTML document as a tree of objects. JavaScript can use DOM APIs to find elements, read or change content, create and remove nodes and respond to user interactions. The DOM is not a database or a CSS layout system.

4. A — addEventListener()

Explanation: addEventListener() registers a function to run when a specified event occurs on an event target. It allows multiple listeners to be registered for the same event without replacing one another. Event types include click, input, change, submit and keydown.

5. A — The element on which the listener is registered

Explanation: event.currentTarget refers to the element on which the currently executing event listener was registered. This differs from event.target, which identifies the object where the event originated and may be a descendant of currentTarget because events can bubble through the DOM. The distinction is important when one listener handles events from several nested elements.

6. D — "Salad"

Explanation: Array indices start at 0, so items[0] is "Soup", items[1] is "Salad" and items[2] is "Pasta". The length of this array is 3. Confusing an item's everyday position with its zero-based index is a common source of mistakes.

7. A — filter()

Explanation: filter() calls a test function for the array's existing elements and returns a new array containing those for which the test is truthy. For example, prices.filter(price => price < 10) selects cheaper items. It does not itself remove the rejected values from the original array.

8. C — The first matching element in document order

Explanation: querySelector returns the first element that matches the CSS selector, or null when no element matches. To obtain all matches, use querySelectorAll, which returns a static NodeList. Code should account for a possible null result before trying to change the returned element.

9. B — textContent

Explanation: Assigning textContent replaces the node's contents with text. Characters such as < and > remain text instead of being interpreted as markup. innerHTML parses HTML and has different behaviour; textContent is suitable when the intended output is a message rather than HTML structure.

10. A — Cancels the event's default form-submission action

Explanation: `preventDefault()` cancels the default action of a cancelable event, allowing a submit handler to perform its own processing without ordinary form submission. It does not remove other listeners or stop propagation. `stopPropagation()` controls propagation, which is a separate part of event handling.